

FLIGHT OPERATIONS

AUDIT PROGRAM #50-10-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
June 3, 2025	Planning	Audit Program Risk Alignment	Joana Leal	Changed defined risk levels
January 30, 2025			Doreen Glowienka	Some minor updates were made by the process owner, updating systems names and wording.
January 30, 2025		Planning	Joana Leal	Changes to audit program risk alignment
January 30, 2025	2	Process Definition and Objective - Process	Joana Leal	Updated Enterprise Risk Management
July 5, 2023	2	Process Definition and Objective - Process	Joana Leal	Updated Enterprise Risk Management
July 5, 2023	3	Reference Material	Joana Leal	Updated links
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

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PROCESS AND OBJECTIVE

For the purposes of this audit program, flight operations is defined as the entire process of providing air transportation on IBM-owned, leased, and chartered aircraft in a safe and cost effective manner. Included in the process are administrative procedures which ensure pilot health and safety, aircraft safety, asset security, aircraft maintenance, and selection of charter companies. Additionally, financial procedures to procure, store, and account for the goods and services are required for continued, safe operations. Lastly, the process includes controls to comply with Federal Aviation Administration (FAA) and other governmental agency regulations.

Enterprise Risk Management:

<i>Strategic Risks</i>
AI differentiation, leadership, and monetization
Market preference for IBM hybrid multi-cloud offerings and as-a-service offerings
Ecosystem adoption of IBM technologies
Capital allocation strategy effectiveness
Transformation speed and achievement of productivity objectives

<i>Operational, Financial, & Regulatory Risks</i>
Hiring top talent and workforce re-skilling for strategy execution
Cybersecurity threats
Geopolitical tensions
Government and industry regulations
Reputational harm
Resiliency to business interruptions
Shifting economic and financial conditions
Supply chain vulnerability

Not all of these risks are captured by or relate to every process within IBM. The intent is to provide a general guide on which ERM risks generally relate to this process.

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PROCESS AND OBJECTIVE (Continued)

Enterprise Risk Management	Application of ERM	Scope Area Covered
Geopolitics and widening government and industry regulations	The audit team should review policies regarding pilot health and safety, as well as aircraft maintenance, to ensure that passengers and pilots are not exposed to danger, which could result in civil litigation and negatively impact IBM's reputation.	Health and safety Pilot Certifications and Safety
Shifting economic and financial conditions	The audit team should verify the management of inventory parts which could result in increased costs due to undetected inventory losses and controls over hanger access to prevent loss.	Accounting Physical Security
Reputational harm	The audit team should review policies regarding pilot health and safety, as well as aircraft maintenance, to ensure that passengers and pilots are not exposed to danger, which could result in civil litigation and negatively impact IBM's reputation.	Health and safety Physical Security
Preparedness for business interruptions	The audit teams should ensure leadership, process interdependencies, data centers and critical suppliers has a business continuity plan in case of any business interruptions.	Health and safety Pilot Certifications and Safety
Cybersecurity	Management team must always ensure the protection of IBM's information. Furthermore, all auditors should be aware, throughout the audit, that all IBM financial data is protected until the time it is made a matter of public record. This should be done by encrypting or password protecting files and attachments in communications, especially with third-party organizations.	Accounting

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PLANNING: Audit Program Risk Alignment

Flight Operations

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Health and Safety Compliance: Government and Industry	Medium	Health and Safety	All
Inventory Control	Medium	Asset Security	All
Asset Security: Inventory and Physical Security	Low	Asset Security	All
Financial Management: Budgeting, Purchasing, Cost Recovery, and Expenses	Low	Accounting	AC001.S, AC002.P, AC003.S, AC004.P, AC005.S,
Access Management and Data Protection	Low	Separation of Duties	SD001.P
Proper Usage and Reporting of Aircraft	Low	Accounting	AC006.P

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PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
TEAs expenses for flight operations	Listing of all TEAs by employee during the period.	Unusually high expenses or large number of TEAs.	These items could indicate unauthorized travel or fraudulent activities.
Expenses for goods and services	Listing of all flight operations related expenses for goods and services.	Unusual expenses or large purchases.	These items could indicate unauthorized purchases or fraudulent activities.
MySky logs	Review the MySky logs to ensure that all flights are logged.	Pilots flying more than the maximum flight hours per day/month. Non-adherence business continuity guidelines when organizing large group travel.	Inappropriate usage of aircraft for unauthorized flights or pilots not in compliance with government regulations regarding maximum flight hours allowed. Impact to business continuity in the event of accident or incident.
Maintenance Logs	Review the logs to ensure that aircraft are maintained in compliance with regulatory guidelines.	Aircraft are not maintained as required.	Aircraft which are not maintained as required, could be a safety hazard for pilots, passengers, and the general public. Additionally, there could be fines from regulatory agencies.

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PLANNING: Reference Material

Corporate Instruction ENV 203 - Management of Chemicals and Chemical Products
(<https://w3.ibm.com/ibm/corporate-documents/env/env-203>)

Corporate Instruction PROC 167 – Supplier sourced goods and services
(<https://w3.ibm.com/ibm/corporate-documents/proc/proc-167>)

Corporate Instruction Legal 110 – The IBM Security Program (<https://w3.ibm.com/ibm/corporate-documents/leg/leg-110>)

IBM Business Conduct Guidelines (<https://w3.ibm.com/ibm/corporate-documents/business-conduct-guidelines>)

Federal Aviation Administration (FAA) Guidelines
(http://www.faa.gov/regulations_policies/faa_regulations/)

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PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
An individual with responsibilities which would permit the initiation and concealment of irregular activity, without detection, could result in loss to IBM. These irregular activities include the following: personal usage of aircraft without business justification or proper approvals, procurement of supplies or tracking of inventory without proper oversight or approvals, collusion of flight operations staff with federal safety inspectors to bypass safety or maintenance checks.	Adequate separation of duties controls should minimize the risk of fraud.	Review the responsibilities assigned in the areas of operating flights, purchasing, inventory management, and maintenance/inspection for possible separation of duties conflicts. Verify that the following tasks are not performed by the same individual: personnel operating flights and approving flight usage, purchasers ordering and verifying receipt; individual tracking the inventory and performing the inventory verification; and repairs requested, performed and verified by the same individual.

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Objective:

Are pilot health and safety, and aircraft safety requirements being followed?

Risk/Exposure:

Pilots and passengers are exposed to danger, and IBM could be liable for civil or criminal litigation, if the management control system does not effectively ensure compliance with pilot health and safety, and aircraft safety requirements.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Primary Test: Safety (HS001.P)

Objective:

To ensure aircraft are operated in a safe manner and in compliance with regulatory agency guidelines.

Approach:

Review flight dispatch regulations. Test a sample of pilot flight assignments to ensure maximum flight hours per day/month for the pilots are not exceeded without prior review and approval. Review the annual line check performed by the chief pilot of all other pilots and verify that pilots followed standard operating procedures and abided by all safety guidelines.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Primary Test: Charters (HS002.P)

Objective:

To ensure chartered and leased aircraft are obtained from reputable company that has been previously inspected and approved.

Approach:

Review procedures for chartering third party aircraft. Test a sample of chartered aircraft to ensure third party charter companies have been approved prior to use, reviewed annually, and have periodic spot checks performed. Validate the accuracy of payments made for charters.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Primary Test: Emergency Program (HS003.P)

Objective:

To ensure emergency procedures have been developed, are periodically tested, and are followed when emergency situations occur.

Approach:

Review emergency procedures for completeness. Verify that the emergency procedures have been tested. Obtain a list of all emergencies reported during the audit period and verify that the required procedures had been followed.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Primary Test: Maintenance (HS004.P)

Objective:

To ensure aircraft maintenance is performed regularly and inspected in accordance with IBM and regulatory guidelines.

Approach:

Review maintenance procedures on planning, scheduling, performing, and inspecting aircraft and parts for completeness.

From a sample of scheduled and unscheduled maintenance, verify maintenance records indicate service was performed and inspected, records contain an airworthiness stamp returning aircraft to operations as required by regulatory guidelines, confirm inventory records support parts required were used, and if possible, visually verify the maintenance was performed.

For a sample of required inspections verify the inspections were conducted as required by the number of landings or aircraft flight hours. Obtain a list of all current revisions to the maintenance manuals and verify that all manuals are up to date.

Test a sample of shift turnover logs and verify that required communication took place as shift changes occurred.

For tools requiring calibration, test the calibration frequency in order to comply with the tool specifications.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Primary Test: Pilot Training and Medical Certification (HS005.P)

Objective:

To ensure pilots are properly trained for the aircraft they operate and that they are medically qualified to operate aircraft.

Approach:

Review training and medical certification requirements.

Test a sample of pilot records for documentation of training for the aircraft they operate and proper medical certification by IBM and the regulatory agency.

Validate all flight crew members are current with safety training (i.e., MedAire defibrillator training) required by IBM and regulatory guidelines.

Validate that pilot training and certification currency information agree with hardcopy personnel files.

Review monthly currency reports to determine any fallouts of currency by pilots.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Health and Safety

Primary Test: Chemical Safety (HS006.P)

Objective:

To ensure chemicals used in fueling, operating, cleaning, and maintaining the aircraft are properly stored, labeled, handled, and disposed of when no longer required.

Approach:

Test the storage, labeling, handling, and disposition of chemicals used in fueling, operating, cleaning, and maintaining the aircraft to see that they are controlled according to IBM and regulatory policies and procedures, and that periodic reviews of chemical handling and storage are performed. Test that Material Safety Data Sheets (MSDS and SDS) are retained for all chemicals and the chemicals used are on the RESO approved chemical list.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Asset Security

Objective:

Are there controls over IBM's assets to ensure they are accounted for and are only accessible to authorized individuals with a business need?

Risk/Exposure:

Access to IBM assets by unauthorized individuals and failure to account for all assets can result in undetected losses.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Asset Security

Primary Test: Inventory Controls (AS001.P)

Objective:

To ensure IBM's inventoriable assets are properly accounted for and controlled.

Approach:

Review the last annual inventory and prior three rotating inventory count results for completeness, timeliness, variance analysis and resolution, and verify that adjusting entries were accurate and properly approved.

Confirm inventories account for assets out for repair.

Perform test counts for a sample of inventoried assets. Flight operations inventory includes aircraft, spare parts, tools, liquor, fuel, automobiles, personal computers, and mobile phones.

Test a sample of monthly variance reports results for completeness and timeliness, and confirm the reason for variances (i.e., used in maintenance, sent out for repair, etc.) within the Inventory Tracking System.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Asset Security

Primary Test: Physical Security (AS002.P)

Objective:

To ensure IBM's assets are only accessible to authorized individuals with a business need.

Approach:

Obtain a current listing of individuals authorized badge access to the flight operations facilities. Verify all individuals have been authorized to access these areas, that the access is periodically reviewed, and that there is a business need for the access. A variety of access levels exist, hanger, parts stockroom, and general building.

Verify security measures are working as designed and that periodic tests are conducted. A test of the security measures should include door alarms and verification that unauthorized access attempts are investigated.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Objective:

Do accounting controls ensure expenditures are approved and justified and cost recovery is timely?

Risk/Exposure:

While Flight Operations does not actually perform accounting functions, their input is the basis for accounting entries. Inaccurate recording of financial or operational transactions or reimbursement of invalid transactions could result in inaccurate financial records and financial loss to IBM.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Secondary Test: Aircraft Purchases and Utilization (AC001.S)

Objective:

To ensure purchases and continued ownership/lease of aircraft are supported by actual use.

Approach:

Verify the accuracy of aircraft utilization tracking and reporting to management. Review the business cases for aircraft purchases/leases and verify current utilization continues to support the business case.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Primary Test: Budgeting (AC002.P)

Objective:

To ensure flight operations expenses are tracked, analyzed, and reported to management.

Approach:

Review the quarterly financial reports and operating budget statements from the audit period to verify the accuracy of the budget and actual expenses and that all variances have been analyzed.

Perform a month-to-month analysis of actual expenses to determine any unusual variances (i.e., fuel charge increases when flight hours decrease).

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Secondary Test: Purchasing (AC003.S)

Objective:

To ensure purchases are approved, reasonable, lowest price available, competitors are treated fairly, and agreements are according to IBM Procurement guidelines.

Approach:

Review a sample of purchases (manual purchase orders and external packages) for proper approvals, supporting documentation, reasonableness, sole source justification if needed, competitive bidding and evaluation analysis if required, and contracts are reviewed and approved by legal.

Verify that any necessary procurement bypass have been obtained on an annual basis.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Primary Test: Travel Expense (AC004.P)

Objective:

To ensure travel expenses are reasonable, comply with IBM guidelines and are properly documented and approved.

Approach:

Review a sample of TEAs for proper supporting documentation, compliance with IBM guidelines, reasonableness, and approval.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Secondary Test: Cost Recovery (AC005.S)

Objective:

To ensure accurate and timely billing for personal aircraft usage and cost recovery billings to the divisions.

Approach:

Review a sample of dispatch logs for personal aircraft usage to validate that the appropriate information is reported to the relative departments in order to receive proper tax treatment. Review a sample of dispatch logs for all aircraft usage to determine whether divisions were accurately billed for usage by employees for business purposes.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Accounting

Primary Test: Appropriate Usage of Aircraft (AC006.P)

Objective:

Review the usage of aircraft to determine that usage by personnel has been authorized, recorded properly, and aligns with business continuity guidelines.

Approach:

Perform a comprehensive review of aircraft usage within the audit period, by reviewing the Flight Operations Systems Logs and comparing this to the list of personnel authorized to fly on corporate aircraft. This is not focusing on the usage of corporate aircraft by officers, former CEOs, or employees in sensitive positions, as that is reviewed as part of the Officers Expense audit.

Test a sample of trips with large groups to ensure flight operations business continuity guidelines are followed.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Separation of Duties

Objective:

To ensure responsibilities are segregated such that no one individual has responsibilities that would allow them to circumvent the system of controls. This must be reflected in the Separation of Duties analysis.

Risk/Exposure:

An individual with responsibilities which would permit the initiation and concealment of irregular activity, without detection, would cause an exposure to IBM.

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CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Separation of Duties

Primary Test: Separation of Duties (SD001.P)

Objective:

To ensure no one individual has responsibilities that would allow the system of internal controls to be circumvented. All tasks, roles, and users must be indicated in the Separation of Duties (SOD) analysis. Any conflicts and associated secondary controls to mitigate these conflicts must be included in the SOD analysis.

Approach:

Review the responsibilities assigned in the areas of purchasing, inventory management and maintenance/inspection for possible separation of duties exposures. Some potential problem areas are: purchasers ordering and verifying receipt; individual responsible for tracking the inventory is performing the inventory verification; and repairs requested, performed and verified by the same individual.